

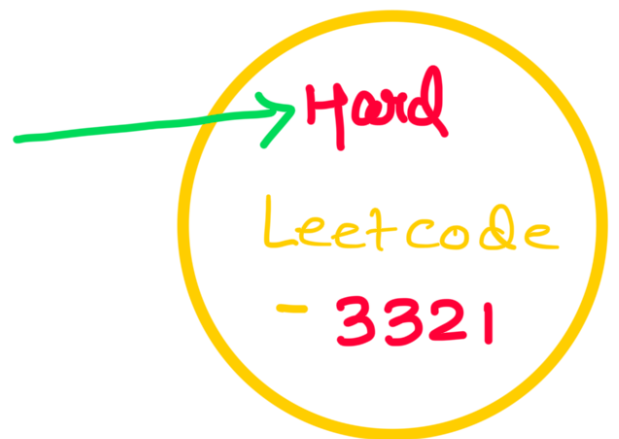
SLIDING

Video - 45



WINDOW

MECHANISM...



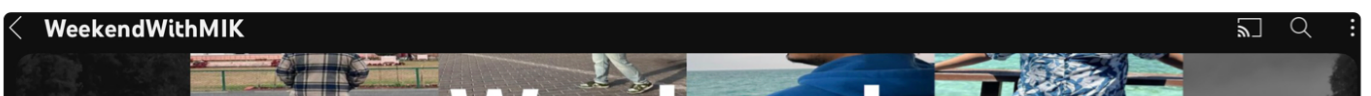
codestorywithmiK



Cs withMIK



codestorywithMIK





Try this channel to
see "Life behind the Scenes"
+
Tech News

Motivation :-

X Something seems difficult because
you already give up in the
beginning.



MIK...

If you stay calm, read it, try it, you will
realise something can be done.

3321. Find X-Sum of All K-Long Subarrays II

Hard

Topics

Companies

Hint

You are given an array `nums` of `n` integers and two integers `k` and `x`.

The **x-sum** of an array is calculated by the following procedure:

$$1 \leq n \leq 10^5$$

- Count the occurrences of all elements in the array.

1 ≤ x ≤ k ≤ n

- Keep only the occurrences of the top x most frequent elements. If two elements have the same number of occurrences, the element with the **bigger** value is considered more frequent.
- Calculate the sum of the resulting array.

Note that if an array has less than x distinct elements, its **x-sum** is the sum of the array.

Return an integer array `answer` of length $n - k + 1$ where `answer[i]` is the **x-sum** of the subarray `nums[i..i + k - 1]`.

Example :- `nums = [1, 1, 2, 2, 3, 4, 2, 3]`

$$k = 6$$

$$x = 2$$

$$\begin{array}{l} 1 \rightarrow 1 \\ 2 \rightarrow 3 \rightarrow 6 \\ 3 \rightarrow 1 \\ 4 \rightarrow 1 \rightarrow 4 \\ \hline 10 \end{array}$$

Output: `[6, 10, 12]`

Thought Process

nums = [1, 1, 2, 2, 3, 4, 2, 3]
 $O(n)$ Sliding w

K = 6

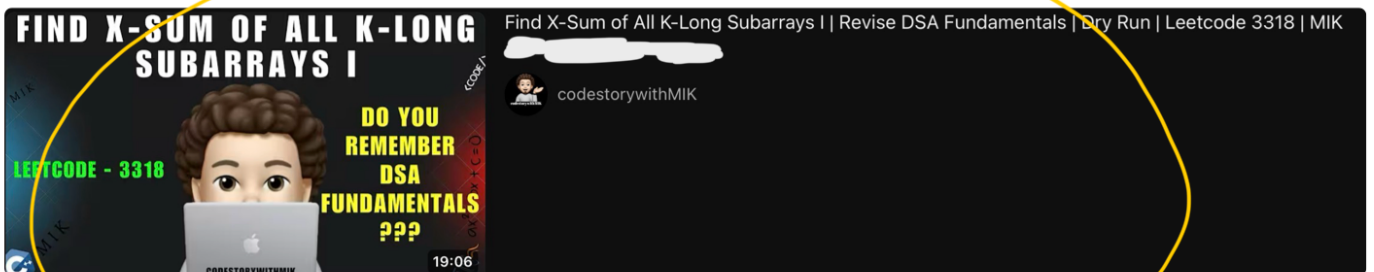
x = 2

$K \log K$

(3, 2)
(1, 3)
(1, 4)
(1, 1)

pq

map
func(map, x)
<freq, elem>



① Sliding window $\rightarrow$ subarray $\rightarrow O(n)$

Improvement in func $\rightarrow$ top-x element $\Leftarrow$

nums = [~~1~~, 1, 2, 2, 3, 4, 2, 3]

K = 6

x = 2

map
1 → 1
2 → 3
3 → 1
4 → 1

~~1, 3~~
~~1, 4~~
1, 1
3, 2

pq



Data Structure → ordered fashion.
<frequency, element>

+ Search the element easily.

Ordered set →

set <<freq, element>>

Will Set Data Structure Help me ???

nums = [1, 1, 2, 2, 3, 4, 2, 3]
 $O(n)$

K = 6

x = 2

$\langle 2, 1 \rangle$
 $\langle \text{freq}, \text{element} \rangle$

map	
value	frequency
1	2
2	3
3	1
4	1

set { (1, 1), (1, 3), (1, 4), (3, 2) } $O(x)$

set.find(2, 2) $\rightarrow \log(n)$

Sum $\rightarrow O(x)$.

$n * (x)$
 $10^5 \times 10^5$

Find Sum Optimally

Set $\{(1,1), (1,3), (1,4), (3,2)\}$

$$O(\log(x)) \ll \cancel{O(x)}$$

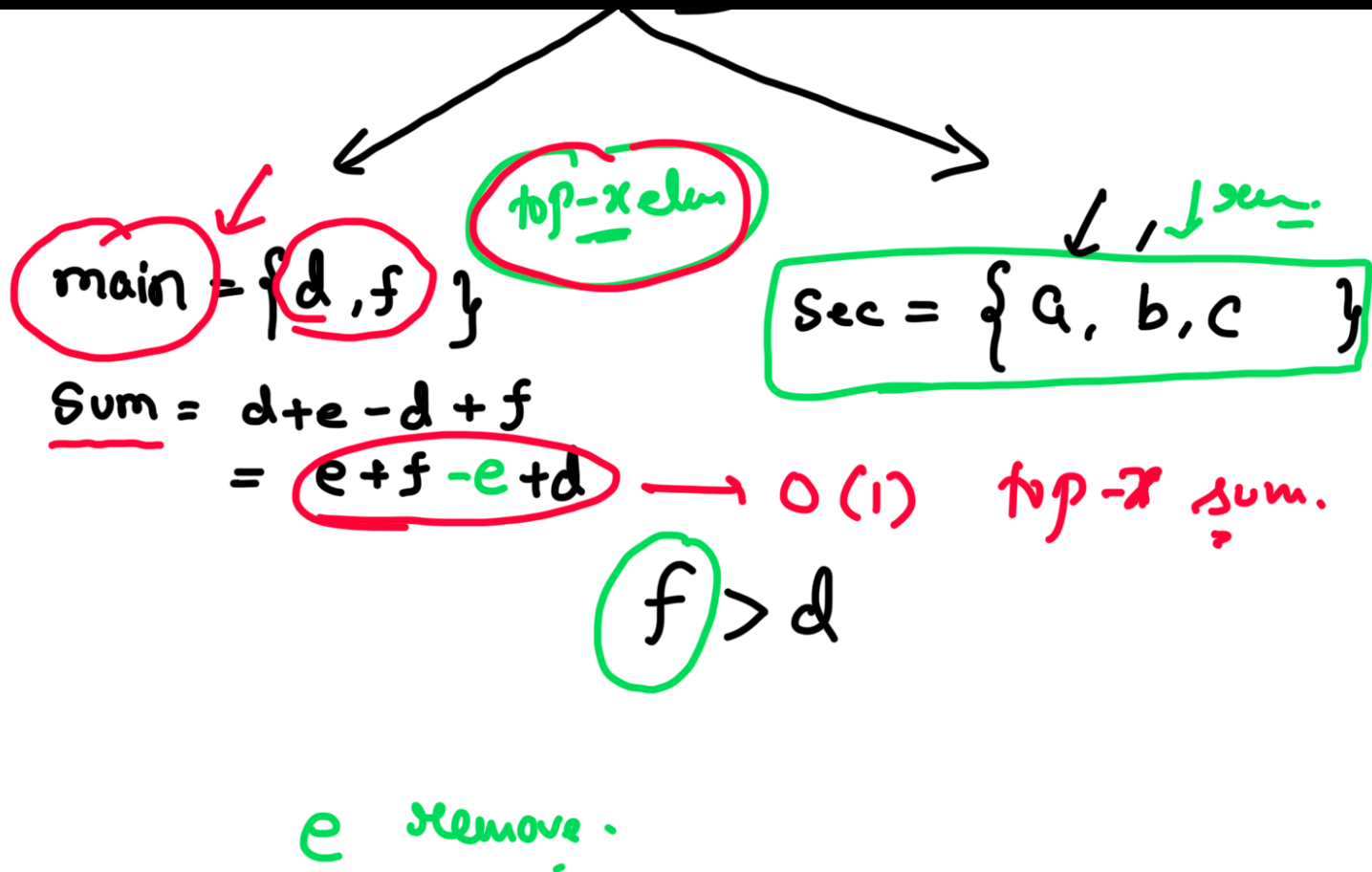
$$\text{Sum} = \text{constant} \prec O(1)$$

$$\text{Sum} = 0$$

* Find the median of a stream:-

$$x = 2$$

Set = $\{a, b, c, d, e\}$
~~Keeping 2 sets.~~



Complete Dry Run.

nums = [1, 1, 2, 2, 3, 4, 2, 3];

K = 6

x = 2

(2, 2)

map	
value	frequency
2	3
3	2

main = $\{(2, 2), (2, 3)\}$

$$\text{Sec} = \{ (1, 4) \}$$
$$\text{Sum} = 4 + 2 - 2 + 4 = 8 - 4 = 4$$
$$+3 = 7 - 3 = 4$$
$$+6 = 10$$
$$10 - 4 = 6 + 6 = 12$$
$$12 - 6 = 6 + 4$$
$$= 10$$
$$10 - 4 = 6 + 4 = 10$$

result = { 6, 10, 12 }

 $O(1)$

(*) Sliding window.

(*) main set $\rightarrow$ x maintained size
 $\quad \quad \quad \searrow$ sum maintained

(*) $\text{sec} \rightarrow \text{remaining } \langle \text{freq, elem} \rangle$



median



Pg . Pg

